The DICTO Ogdoecontahedron on the Octet Truss

Exact space-filling kits: golden keys, a Penrose link between facing five-fold stars, golden envelopes in the octet, and one-piece glue with Ogdoecontahedron imprints

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Version 5 · 5 November 2026 (adds a pointer to Pentagons for the Kites; Version 4 redrew the right panel of Figure 3; Version 3 credited DICTO's ideas as his proposals) · DICTO DOI: 10.5281/zenodo.23457016 · Licence CC BY-SA 4.0 · Data, scripts, exact checks and renders are included with this record and at https://github.com/DICTOR-Master/kaleidohedra (branch research: studies/2026-10-18-ocd-octet, -ocd-octahedron-caps, -p1-golden-key-octet, -p2-penrose-core-octet, -p3-golden-envelope-octet, -p4-golden-octet-adaptor, -envelope-network).
The octet truss built from the Ogdoecontahedron's own pieces, the Penrose link between facing stars and the shared middle piece, the move to a minimal golden kit, the one-piece and two-shape golden octets, the glue pieces with exact Ogdoecontahedron imprints and centred on the Penrose star, the keyed glue, the half-RD network and the chamber are DICTO's ideas and choices. Computation, exact arithmetic and renders were done with Claude (Anthropic) as a research assistant; the independent exact proofs, the vertex glue W, the identification of the tetartoid and the keyed-glue design are by Claude Fable, with DICTO's go.

DRAFT, Version 5, for DICTO's reading. Not published. Delete this box at publication. Version 5 adds a pointer to Pentagons for the Kites (#95): one sentence at the end of §8 and one reference; nothing else changed.

Abstract. Joined hexagon to hexagon, copies of the DICTO Ogdoecontahedron (OG) form a body-centred cubic packing whose gap is treated in a companion paper (DICTO DOI: 10.5281/zenodo.23451210). Here the OG goes the other way: its eight joint hexagons lie exactly on the faces of a regular octahedron $|x| + |y| + |z| = 2\varphi/3$, so the OG sits in the octahedra of the octet truss, and the space round it becomes tractable. Exactly, in $\mathbb{Q}(\sqrt{5})$, we give four families of space-filling kits. (1) A golden-key octet: the OG's golden envelope (a rhombic triacontahedron) pokes into each neighbouring tetrahedron as a small key whose base corners sit at the golden-section points of the octahedron's edges; keyed octahedron and dented tetrahedron (a left and a right) fill space. (2) A Penrose-core octet: the five-fold stars of neighbouring OGs face each other across every octahedron edge, offset by exactly one star tip radius, and an exact thick Penrose rhombus, with two φ-smaller ones, joins them; the shared middle piece holds three Penrose cores. (3) Golden envelopes on every octahedron never overlap and fill exactly $90 - 40\sqrt{5} = 10/\varphi^6 \approx 55.73\%$ of space; the rest is one chiral piece, so envelope and grown piece fill space with two shapes, and each envelope lies inside its rhombic dodecahedron touching all 12 faces at golden-section points. (4) One-piece glue: the rhombic dodecahedron round each octet vertex, or the octahedral cell round each edge midpoint, minus the OGs that bite it, fills space with the OG as two shapes with no mirror pair, the sockets being exact OG imprints; the edge-centred piece holds all three Penrose cores and can be keyed with the OG's own kites so that twelve copies lock round one OG. The main kits were proved exactly by an independent engine and checked with millions of sample points.

1. The OG and the octet frame

The DICTO Ogdoecontahedron (OG) is a convex solid of 20 regular hexagons and 60 kites, the polar of the DICTO Hecatodihedron, exact in every coordinate (DICTO DOI: 10.5281/zenodo.23404688). In its bcc packing it touches eight neighbours through eight hexagons; the gap of that packing, 22.57% of space, is the subject of the companion paper Golden Bodhi Pyrochlore (DICTO DOI: 10.5281/zenodo.23451210), where DICTO's one-piece answer is the Bull-horns OG adaptor. This paper follows a second route, which DICTO opened by asking for "an octet truss system from pieces available".

The OG in its octahedron (exact). The OG's eight joint hexagons lie exactly on the faces of the regular octahedron $|x| + |y| + |z| = R$ with $\mathbf{R = 2\varphi/3}$ (edge $2\sqrt{2}\varphi/3 \approx 1.5255$); every corner of the OG is inside or on it, 48 of them on it. The OG, of volume $(-1000 + 1240\sqrt{5})/1539 \approx 1.151868$, fills 68.83% of its octahedron. The six corner cones cut the rest into six equal caps, $(1108 - 316\sqrt{5})/4617 \approx 0.086940$ each.

The octet frame (exact). Put the octahedra on the face-centred cubic lattice of vectors R(1, 1, 0) and its turns, and the regular tetrahedra $T+ = \mathrm{conv}(\mathrm{Re}_x, \mathrm{Re}_y, \mathrm{Re}_z, R(1, 1, 1))$ and $T- = -T+$ between them: the octet truss of Fuller (1961), the tetrahedral–octahedral honeycomb. One octahedron and two tetrahedra make one lattice cell, $(64 + 32\sqrt{5})/81 + 2 \cdot (16 + 8\sqrt{5})/81 = (32 + 16\sqrt{5})/27 \approx 2.510263$. Every face of T+ lies on an octahedron face of one tetrahedral class and every face of T− on the other, so each OG meets four T+ on one tetrahedral set of joint hexagons and four T− on the other. The plain kit, one OG, six caps, T+ and T−, already fills space; the OGs alone take $(1025 - 435\sqrt{5})/114 \approx 45.886\%$ of it.

What "exact" means here. All geometry is in $\mathbb{Q}(\sqrt{5})$, as pairs of rationals. Corners are found by solving plane triples (a float pre-screen may only discard a triple whose float solution misses a plane by more than 10^{-7}) and every corner is checked exactly against every plane; volumes come from exact corners. The kits marked "proved" in Table 1 were proved to fill space exactly once by Claude Fable with an engine written independently of the study's, without symmetry shortcuts, and checked on dicto-node with at least 4 000 000 sample points per kit (uniform and near-face): no point uncovered, none covered twice. Meshes (manifold3d) confirm that each piece is one solid. Quantities marked float, such as the sphericity $\psi = \pi^{1/3}(6V)^{2/3}/A$ (1 for a ball, 0.806 for a cube), are not exact unless stated.

2. Kit 1: the golden-key octet (exact)

The key. The OG's golden envelope, the rhombic triacontahedron (RT) of inradius 2/3 in the same frame, pokes through each joint plane as a small tetrahedron: its three-fold corner, at $|x| + |y| + |z| = 2/\varphi$, over three RT corners that lie exactly on the plane. Those three base corners lie **on the octahedron's edges, at their golden-section points (φ : 1)**, so the key's base triangle is inscribed in the octahedron face. Each key, of volume $(-44 + 20\sqrt{5})/81 \approx 0.008906$, lies wholly inside its tetrahedron, and its three side faces are exact half golden rhombi (31.72°, 31.72°, 116.57°). The four keys entering a tetrahedron never meet.

The kit. All eight keys go out from the OG as bumps and each tetrahedron takes four dents. Keyed OG $(-7688 + 4280\sqrt{5})/1539 \approx 1.223113$, six caps, and two dented tetrahedra $64/27 - 8\sqrt{5}/9 \approx 0.382754$ make the cell exactly. Fusing the OG and its caps gives the **golden-key octet**: the keyed octahedron, $-32/9 + 64\sqrt{5}/27 \approx 1.744754$ (69.50% of space, ψ 0.852), and the dented tetrahedron (15.25% each, ψ 0.620; Figure 1). The dented tetrahedron is chiral:

because the keys' base corners sit at golden-section points, each key is turned, the piece has the 12 rotations of the tetrahedron and no reflection, and T− is the mirror image of T+ (an earlier reading in the study that they were the same piece was corrected by Fable). So the kit is two shapes, one of them in a left and a right version.



Figure 1. The golden-key octet: the keyed octahedron (the OG with its caps, gold, and its eight golden-envelope keys) and the dented tetrahedron in its left (green) and right (teal) versions; eight tetrahedra round one octahedron; the network.

3. Kit 2: facing stars and the Penrose-core octet (exact)

Facing stars. Each of the OG's twelve five-fold corners carries a star of five kites, and each faces one octahedron edge. Across the edge it meets the neighbouring OG's corner c2 = R(0, 1, 1) − c1, pointing straight back. Seen down the five-fold axis (Figure 2), **the sideways offset of c2 equals the star's tip radius, 0.350487, exactly**; c2 sits in line with an inner corner of the first star, between two tips; the two stars are turned a half turn (36° for a five-fold star) against each other; and along the axis c2 is 0.0943 nearer. All of this is fixed by the lattice, so alignment has to come from a joining piece. The half-turn centre m = R(0, 1, 1)/2 is the midpoint of the octahedron edge the two OGs share and a centre of inversion that swaps them.

The Penrose link. DICTO proposed that the five-lobed patterns could become Penrose prisms that connect the OGs. Down the axis, c1 → c2 is exactly the long diagonal of a **thick Penrose rhombus** (72° at c1 and c2, 108° at the sides) of edge tip radius ÷ φ ≈ 0.2166, whose edges run along the stars' own spokes; at each side corner a thick Penrose rhombus one φ-step smaller (edge ≈ 0.1339) reaches one tip of each star. Three thick Penrose rhombi in φ-scaled sizes thus join the two stars tip to tip, symmetric about m.

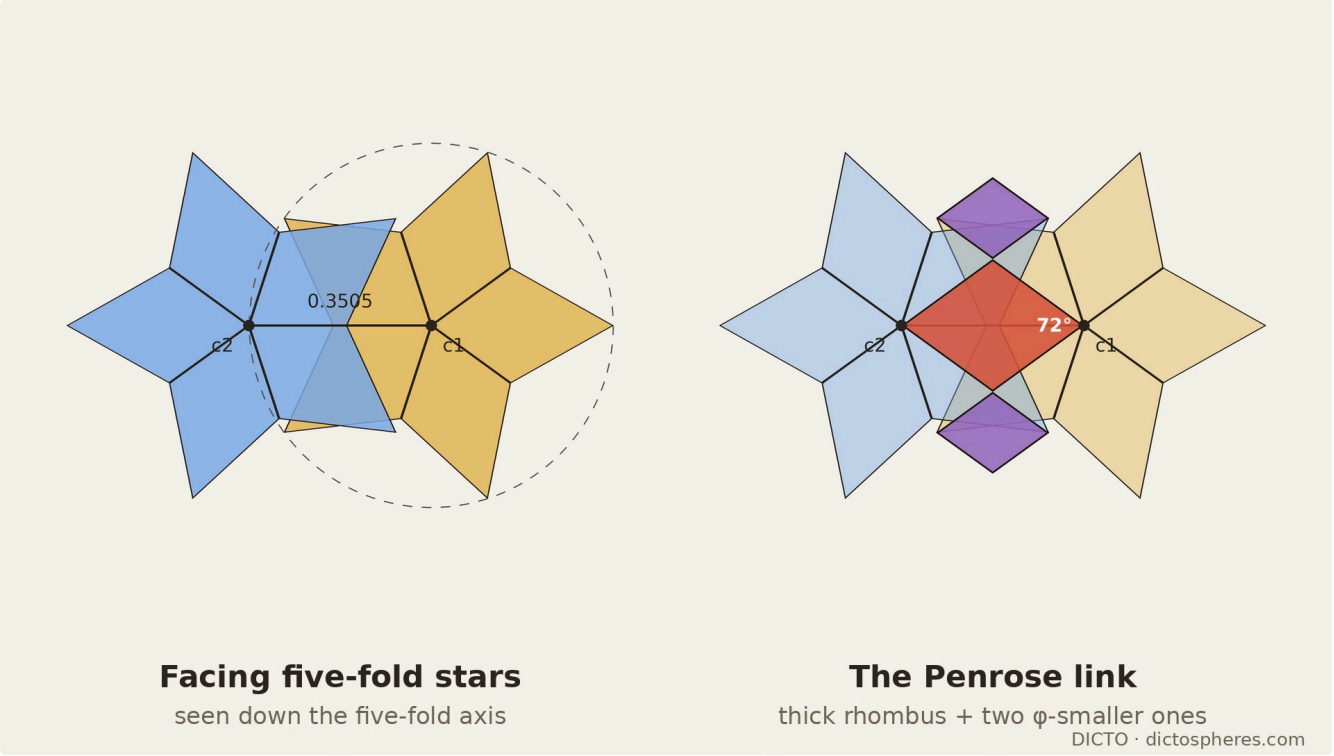


Figure 2. Down a five-fold axis: the stars of two neighbouring OGs (gold round c1, blue round c2), offset by exactly the tip radius (dashed circle), and the Penrose link: a thick Penrose rhombus with long diagonal c1c2 (red) and two φ-smaller thick rhombi (purple).

The middle piece and its cores. DICTO chose a shared middle piece on the half-turn centre. Let H be the hull of the two facing stars, an oblique pentagonal antiprism of volume $4\sqrt{5}/81$; H stays within the two octahedra and two tetrahedra round its edge, and no two hulls overlap. The middle piece is M_0 = H minus both OGs, $(310 + 26\sqrt{5})/4617 \approx 0.079735$, one per octahedron edge (six per cell). The three Penrose prisms, each run along the axis from one OG's surface to the other's and trimmed to H, give three cores, pairwise disjoint: the central core $(-36734 + 16658\sqrt{5})/50787 \approx 0.010129$ and two flanking cores $4114/1539 - 9182\sqrt{5}/7695 \approx 0.004993$ each, a mirror pair (no rotation takes one to the other). Together they are 25.2% of M_0; the rest is $-231380/50787 + 524152\sqrt{5}/253935 \approx 0.059620$.

The kits. With the keys of §2 the middle pieces are trimmed round four key tips; Fable proved both the keyed and the full kit exactly, and suggested dropping the keys, which DICTO chose. The **Penrose-core octet** as a minimal set fuses the OG with its notched caps: notched octahedron $8/57 + 104\sqrt{5}/171 \approx 1.500299$ (59.77% of space), middle piece M_0 (six per cell, 19.06% in all), and notched tetrahedron $26/81 - 2\sqrt{5}/81 \approx 0.265776$ (a left and a right, 21.18% together). With the three cores cut out (DICTO's choice), the keyless kit is OG, notched cap, central core, flanking core (left and right), rest and tetrahedron (left and right): seven shapes, five counting mirror pairs as one. Per cell: $1.151868 + 6 \times 0.058072 + 6 \times 0.079735 + 2 \times 0.265776$ = the cell, exactly. In the keyed kit the thinnest pieces, the middle rests and the caps, were also tried merged into vertex or tetrahedron pieces; each merge made the pieces less chunky (float), because the thin pieces come from the octet's spacing itself.



Figure 3. The Penrose-core octet: notched octahedron (blue), middle piece (red) and notched tetrahedron, left and right (green, teal); twelve middle pieces and eight tetrahedra round one octahedron; the keyless kit with each middle piece split into its three Penrose cores (purple) and the rest.

4. Kit 3: golden envelopes in the octet (exact)

The envelope network. DICTO proposed developing the octet "toward a minimal golden kit". Put the golden envelope (RT, $(-320 + 160\sqrt{5})/27 \approx 1.398921$) on every octahedron. Neighbouring envelopes never overlap (exact; their closest approach is along facing parallel edges, $2/(3\varphi^3\sqrt{(\varphi\sqrt{5})}) \approx 0.0827$ apart), and each envelope is exactly its part inside its octahedron plus its eight keys of §2. So **the envelopes fill exactly $90 - 40\sqrt{5} = 10/\varphi^6 \approx 55.728\%$ of space**. At every octet vertex the six corner pyramids of the octahedra cut off by the envelopes' face planes, each the octahedron's corner scaled by φ^{-2}, and the φ^{-2} corners of the eight tetrahedra make a **cuboctahedron scaled by φ^{-2}**, which shares a whole rhombus face with each of its six envelopes and has **exactly one third of the envelope's volume**, $(-320 + 160\sqrt{5})/81$, filling 18.576% of space. As DICTO noticed, none of this needs the OG: the network stands on its own.

Two shapes. DICTO proposed growing the small dodecahedral cell at the centre of each tetrahedron into a smaller kit. Giving every point outside the envelopes to its nearest tetrahedron centre, the regions are cubes of edge R (the tetrahedron centres (R/2)(odd, odd, odd) form a cubic lattice), whose corners alternate between four OG centres and four octet vertices. Exactly four envelopes bite each cube, RT/8 each, so the **grown piece G = cube − RT/2 = $(176 - 72\sqrt{5})/27 \approx 0.555671$**. G is one solid with 44 corners, 72 edges and 30 faces (12 whole golden rhombi, 12 quarter rhombi and 6 congruent non-convex decagons, the cube's cuts, which carry golden angles), dihedral angles only 90°, 108° and 216° (reflex), and $\psi = 0.6025$; its symmetry is exactly the rotation group T, so the piece at a T− site is its mirror image. **Golden envelope (with or without the OG inside) and grown piece fill space exactly**: two shapes, one in a left and a right version (Figure 4). At G's centre sits the original dodecahedral cell, which Fable identified exactly as a **tetartoid**: twelve congruent pentagons, symmetry T, volume $(-146 + 66\sqrt{5})/27 \approx 0.058537$, edge lengths ≈ 0.157 ($= R/\varphi^4$), 0.242 and 0.299.

The envelope inside its rhombic dodecahedron. Giving every point instead to its nearest envelope centre, the region is the fcc Voronoi cell, the rhombic dodecahedron $|x| + |y|, |y| + |z|, |z| + |x| \le R$. **The envelope lies inside it and touches each of its 12 faces at one corner**, the 12 golden-section points on the octahedron's edges ((2/3, (√5 − 1)/3, 0) and its

turns). The adaptor, rhombic dodecahedron minus envelope, $(352 - 144\sqrt{5})/27 \approx 1.111341 = 2G$ (44.27% of space), is one centrosymmetric solid with the envelope sealed inside; it opens into two identical halves along a coordinate plane, each of G's volume. DICTO's sets: the **one-piece golden octet** (the rhombic dodecahedron with its envelope fused, filling space alone as the classical fcc tiling; Figure 4, right), the **two-shape golden octet** (solid envelope + adaptor half, no mirror images) and the adaptor as a socket for **any solid whose golden envelope this is**, the OG among them.

An intermediate refinement, envelope + φ^{-2} cuboctahedron + small corner wedges + cut tetrahedra, was retired: the envelope touches each octahedron edge at one point only, so the wedges are spikes attached by slivers and cannot be made (Fable). The φ^{-2} facts above stand.

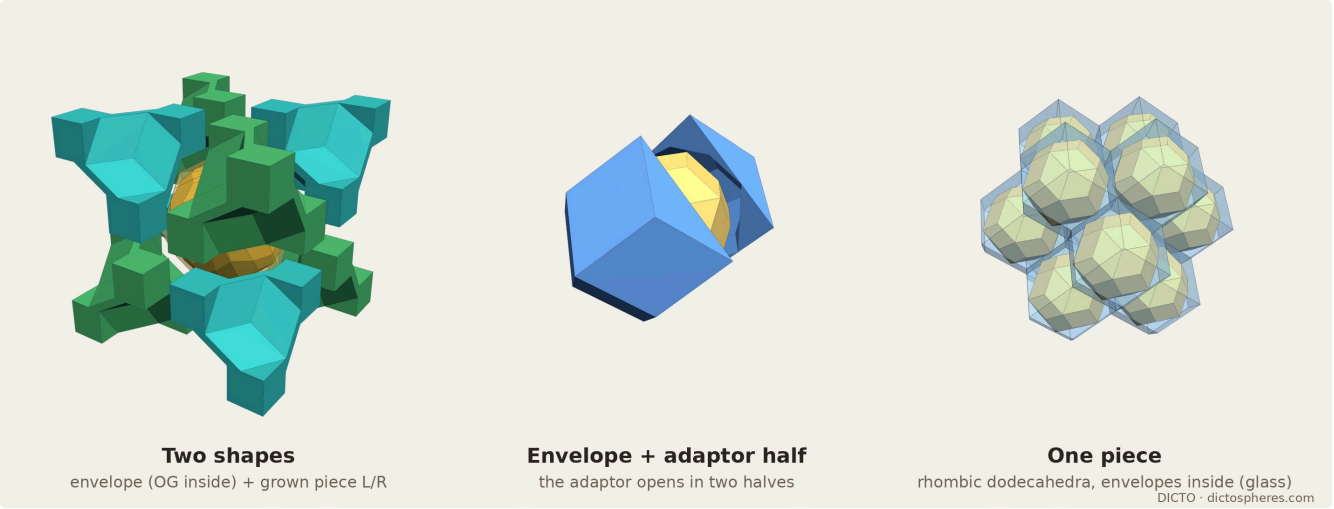


Figure 4. Golden envelopes in the octet: the two-shape kit, envelope (with the OG inside) and grown piece, left (green) and right (teal); the envelope with the two halves of its adaptor (blue); the one-piece golden octet, rhombic dodecahedra with the envelopes inside (glass).

5. Kit 4: one-piece glue with OG imprints (exact)

Envelope + W. Fable proposed a glue piece centred on the octet vertices: W = the rhombic dodecahedron round an octet vertex minus the six envelopes that bite it. W is one solid of volume $(352 - 144\sqrt{5})/27 = 2G$, achiral (symmetry m3), with the φ^{-2} cuboctahedron inside it (ψ 0.574, sharpest edge 77.4°). **Envelope + W fills space exactly with two shapes and no mirror pair.**

OG + W with OG imprints. DICTO proposed that the faces be exact imprints of the OG. W_{OG} = the vertex rhombic dodecahedron minus the six OGs that bite it, each bite exactly OG/6: volume $(2824 - 328\sqrt{5})/1539 \approx 1.358395$ = cell − OG (54.11% of space), one solid of genus 0, symmetry exactly m3. Its 132 faces include 120 imprint faces that together are exactly one whole OG surface. Its sharpest edge is 86.9° (no knife edges), two sockets in one W come no closer than $\sqrt{(70/81 - 10\sqrt{5}/27)} \approx 0.18980$, and $\psi = 0.648$. **OG + W_{OG} fills space exactly: two shapes, no mirror images, the OG whole.**

OG + W centred on the Penrose star. DICTO then proposed centring the imprint on the Penrose star. The Voronoi cell of an octahedron-edge midpoint m among all of them is an octahedron with corners at the two OG centres, the two edge ends and the two tetrahedron centres round that edge, of volume $(16 + 8\sqrt{5})/81$ = cell/6. W_{star} = that cell minus its two OGs (each bite OG/12): $(1412 - 164\sqrt{5})/4617 \approx 0.226399$, one solid of genus 0, symmetry exactly

2/m, 32 faces, sharpest edge 75.4°, ψ = 0.690, the chunkiest glue found. It contains 97.7% of the middle piece of §3 and, exactly, **all three Penrose cores**. Twelve W_{star} surround each OG, six per cell; **OG + W_{star} fills space exactly with two shapes and no mirror pair** (Figure 5).

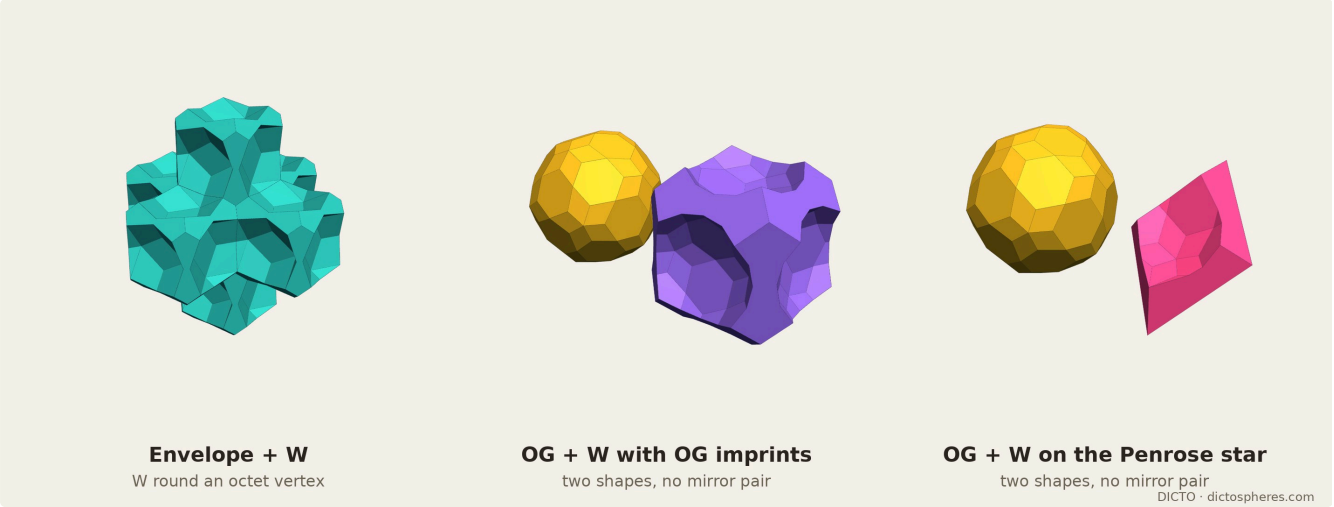


Figure 5. One-piece glue: W round an octet vertex with its six envelope sockets (Fable); the OG with W_{OG}, whose sockets are exact OG imprints; the OG with W_{star}, centred on the Penrose star between two OGs.

The keyed star-centred W. DICTO proposed that the piece key with itself through the star's own pattern, while staying a single piece. The eight flat W–W faces of W_{star} pair with neighbouring faces under the structure's symmetries; no face pairs with itself and the pairing is bipartite (exact), so four faces can carry raised keys and the other four the same keys sunk, with every W still identical. In Fable's design each flat face carries one kite congruent to the OG's star kite (long edge 0.2379, short edge 0.1971), set so that it reads as the star's next kite carried onto the flat face; one kite per face is the most that fits at the star's own scale. The key's position, small clearances and wall slope are design choices made by search (float), and the checks are by meshes and sampling: the keyed W is one solid of genus 0 with W_{star}'s volume; each bump fills its neighbour's dent exactly; OG + keyed W fills space with no gaps or overlaps in 6 000 000 sample points; and of 600 directions tested, **no W in the ring of twelve round an OG can be slid out** (the unkeyed W: 41 free directions). The ring is assembled by placing the W pieces one by one, each pushed on along a bump-face normal (Figure 6).

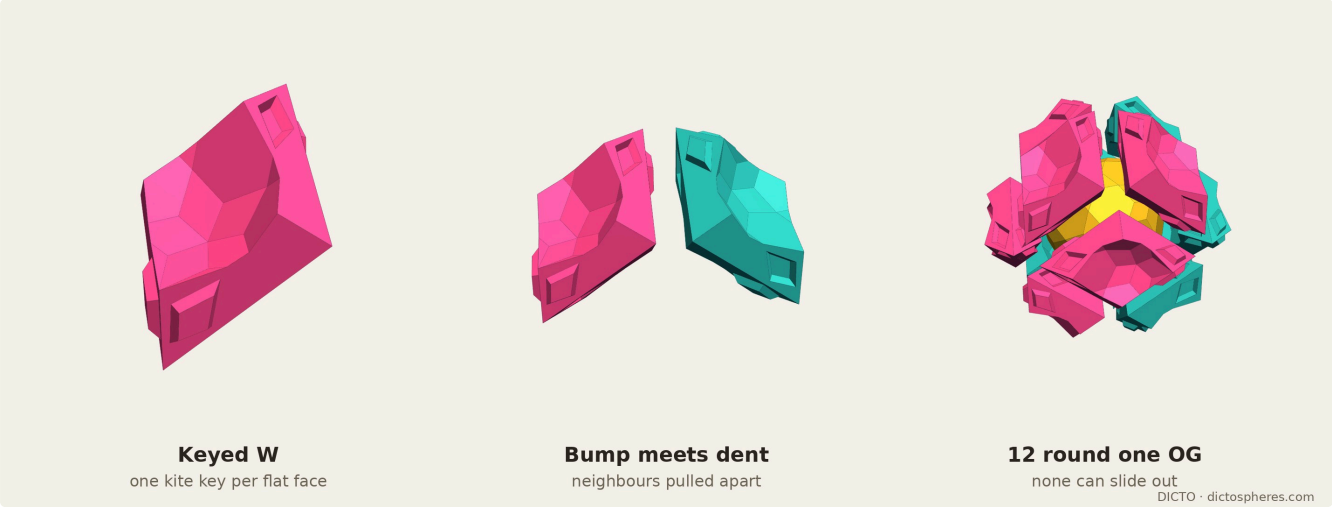


Figure 6. The keyed star-centred W (Fable's design): one piece with a kite key on each flat face; a bump meeting its dent (pulled apart); twelve identical keyed W round one OG.

6. A half-RD network, and a chamber (outside the plan)

DICTO's generalisation: cells equivalent to the rhombic dodecahedron (a golden envelope with its adaptor), each either full or half-open (only one adaptor half kept), arranged in any network on the fcc lattice, with any solid whose golden envelope this is in each socket. Two such cells are known from §4–5: the envelope-centred rhombic dodecahedron (opening in two halves) and the vertex-centred W. One instance, at DICTO's request, is a hollow spherical chamber: the cells with centres between radii 2.4 and 5.4 (230 cells, in the frame of $R \approx 1.0787$), the outer layer of 162 cells carrying only their inward adaptor half and the 60 cells lining the hollow only their outward half, so that the envelopes are exposed outside and inside (Figure 7). This is geometry only; nothing structural is claimed.

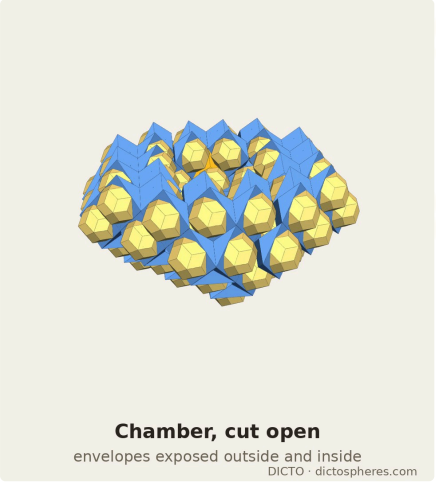


Figure 7. Outside the plan: a hollow chamber of half-open cells, cut open, envelopes exposed on its outer and inner surfaces.

7. The kits at a glance

Kit	Shapes (mirror pairs counted once)	Shares of space	Status
Golden-key octet	keyed octahedron, dented tetrahedron L/R (2)	69.50% / 30.50%	exact, proved
Penrose-core octet	notched octahedron, middle piece, notched tetrahedron L/R (3)	59.77% / 19.06% / 21.18%	exact, proved
Keyless kit, three cores	OG, cap, central core, flanking core L/R, rest, tetrahedron L/R (5)	OG 45.89%	exact, proved
Golden envelope octet	envelope, grown piece G L/R (2)	55.73% / 44.27%	exact, proved
Two-shape golden octet	envelope, adaptor half (2, no mirror pair)	55.73% / 44.27%	exact
Envelope + W	envelope, W (2, no mirror pair)	55.73% / 44.27%	exact, proved
OG + W_{OG}	OG, W_{OG} (2, no mirror pair)	45.89% / 54.11%	exact, proved
OG + W_{star}	OG, W_{star} (2, no mirror pair)	45.89% / 54.11%	exact, proved
OG + keyed W_{star}	OG, keyed W (2, no mirror pair)	45.89% / 54.11%	design; meshes and sampling

Table 1. The kits of this paper. "Proved": an independent exact proof of filling space once (Claude Fable), backed by sampling. Shares are exact numbers, rounded.

8. Relation to known work, and status

Known and credited: the octet truss (Fuller 1961) and the tetrahedral–octahedral honeycomb, including other tilings by tetrahedra and octahedra (Conway, Jiao and Torquato 2011) and part counts of layered octet trusses (Lalvani 1990); the rhombic dodecahedron as the Voronoi cell of the fcc lattice; the rhombic triacontahedron and golden rhombohedra as golden zonohedra; Penrose rhombi (Penrose 1974); the tetartoid as a crystal form with the rotation symmetry of the tetrahedron; Voronoi cuts round chosen sites. Densest packings: the densest known packing of the rhombic triacontahedron is a lattice packing at 0.801741 (Torquato and Jiao 2012; Betke and Henk 2000); our envelope network, at $10/\varphi^6 \approx 0.5573$, is a different and sparser lattice packing, of interest for its exact fit to the octet, not its density.

Closest work: topologically interlocking blocks built from the tetrahedra and octahedra of the same honeycomb (Akpanya, Goertzen and Niemeyer 2024) and interlocking space-filling tiles from Voronoi partitions (Krishnamurthy, Akleman et al. 2022) are close in spirit to the keyed W; neither uses icosahedral sockets or Penrose keys. Lalvani's periodic structures of nodal polyhedra joined by connector polyhedra (1995) are the closest concept to "solid + glue piece", with cubic and tetrahedral symmetries and no golden envelopes; his triacontahedral packing with gaps (1997, Fig. 4D) stacks a non-regular triacontahedron cubically. Kabai's "Golden Octet Truss" (2007) is a different object, a golden rhombohedron cut into a tetrahedral and an octahedral part, and we avoid the name; his demonstration relating the rhombic triacontahedron and the rhombic dodecahedron (2008) shows a five-fold compound of rhombic dodecahedra round a triacontahedron, not one rhombic dodecahedron holding a triacontahedron. A "dimpled" rhombic triacontahedron model that packs on lattices is a modified solid, not the plain one used here. Koskinen's triacontahedra that fill space twice (2011) and Kabai's triacontahedral clusters concern other arrangements.

Not found in our reference library or web searches (18–19 October 2026): rhombic triacontahedra placed in the octahedra of the octet truss, their exact fill $10/\varphi^6$, the golden-section keys on the octahedron edges, the φ^{-2} vertex cuboctahedron of one third of an

envelope, the triacontahedron touching its fcc rhombic dodecahedron at the twelve golden-section points, the Penrose link between facing five-fold stars of neighbouring solids, and glue pieces carrying exact imprints of an icosahedral solid. **Exact:** §1–5 except as marked. **Float:** sphericities other than G's, the merge experiments in §3, the keyed-W design and its locking tests, and the chamber.

Later work. The octet-truss kit for the cut OG, the OG with its twelve five-fold tips cut into pentagons, is given in Pentagons for the Kites (DICTO DOI: 10.5281/zenodo.23968270): four shapes and a left/right pair, built on a pentagonal connector across every octahedron edge.

9. Reproducibility

In studies/2026-10-18-ocd-octet: step1_ocd_in_octahedron.py, step2_octet_frame.py (§1), step3_key_candidate.py, step5_keyed_octet.py (§2), step4_five_fold_stars.py, step13_penrose_cut.py, step11_exact_pieces.py, step15_penrose_core.py, step16_space_shares.py, step17_vertex_piece.py, step18_no_keys_flanks.py, step19_flank_cores.py, step20_keyless_meshes.py (§3), step21_envelopes_in_octet.py, step22_complement.py, step24_outer_corner_cut.py, step26_tet_plane_trick.py, step27_expanded_core.py, step27b_chirality.py (§4), with the exact helpers q5.py and exactpoly.py; Fable's reviews with their own engine and sampling in fable-review/ (keyed and full kits) and fable-review-2/ (two-shape kit, tetartoid, W). In studies/2026-10-18-p1-golden-key-octet, -p2-penrose-core-octet and -p3-golden-envelope-octet: the fused sets (p1_meshes.py, p2_meshes.py, p3c_dodeca.py). In studies/2026-10-18-p4-golden-octet-adaptor: p4_adaptor.py, p4_mesh.py (§4), p4_W.py, p4_W_ocd.py, p4_W_star.py, p4_key_pairing.py (§5), p4_chamber.py (§6); fable-review-3/ (W_{OG} and W_{star}, exact) and fable-review-4/ (the keyed W). Prior-art notes: studies/2026-10-18-envelope-network. Figures: papers/2026-10-18-bull-horns/scripts (shoot_clean.mjs, draw_stars.py, make_figs.py). Every study README records each step, DICTO's words and the corrections.

References

1. DICTO, The DICTO Ogdoecontahedron, DICTO DOI: 10.5281/zenodo.23404688.
2. DICTO, Golden Bodhi Pyrochlore, DICTO DOI: 10.5281/zenodo.23451210.
3. DICTO, Pentagons for the Kites, #95, DICTO DOI: 10.5281/zenodo.23968270.
4. R. Akpanya, T. Goertzen, A. C. Niemeyer, Topologically interlocking blocks inside the tetroctahedrille (2024), arXiv:2405.01944.
5. U. Betke, M. Henk, Densest lattice packings of 3-polytopes, Computational Geometry 16 (2000) 157–186, doi:10.1016/S0925-7721(00)00007-9.
6. J. H. Conway, Y. Jiao, S. Torquato, New family of tilings of three-dimensional Euclidean space by tetrahedra and octahedra, PNAS 108 (2011) 11009–11012, doi:10.1073/pnas.1105594108.
7. R. B. Fuller, Synergetic building construction, US Patent 2,986,241 (1961).
8. S. Kabai, Golden Octet Truss, Wolfram Demonstrations Project (2007), https://demonstrations.wolfram.com/GoldenOctetTruss/.
9. S. Kabai, Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron, Wolfram Demonstrations Project (2008), https://demonstrations.wolfram.com/RelatingARhombicTriacontahedronAndARhombicDodecahedron/.
10. S. Kabai, A Rhombic Dodecahedron Made of Rhombic Triacontahedra, Wolfram Demonstrations Project (2008), https://demonstrations.wolfram.com/ARhombicDodecahedronMadeOfRhombicTriacontahedra/.
11. S. Kabai, Cluster of 14 Rhombic Triacontahedra, Wolfram Demonstrations Project (2015), https://demonstrations.wolfram.com/ClusterOf14RhombicTriacontahedra/.
12. J. Koskinen, Unusual tilings and transformations, Bridges 2011 Proceedings, https://static1.bridgesmathart.org/2011/cdrom/proceedings/118/paper_118.pdf.

13. V. R. Krishnamurthy, E. Akleman et al., Geometrically interlocking space-filling tiling based on fabric weaves, IEEE Transactions on Visualization and Computer Graphics 28(10) (2022), https://midl.tamu.edu/data/papers/tvcg_woven_2021.pdf.
14. J. C. Lagarias, C. Zong, Mysteries in packing regular tetrahedra, Notices of the AMS 59(11) (2012) 1540–1549.
15. H. Lalvani, Transformational part-count in layered octahedral-tetrahedral truss configurations, NASA TM-102635 (1990), https://ntrs.nasa.gov/citations/19900019282.
16. H. Lalvani, Periodic space structures composed of two nodal polyhedra for design applications, US Patent 5,448,868 (1995).
17. H. Lalvani, Building systems with non-regular polyhedra based on subdivisions of zonohedra, US Patent 5,623,790 (1997).
18. R. Penrose, The role of aesthetics in pure and applied mathematical research, Bull. Inst. Math. Appl. 10 (1974) 266–271.
19. S. Torquato, Y. Jiao, Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal (2012), arXiv:1206.3716.
20. Dimpled Rhombic Triacontahedron – Construction, 3D model, Sketchfab, https://www.sketchfab.com/3d-models/dimpled-rhombic-triacontahedron-construction-9e2be3ccfe7b480c846f4d431441ac19 (accessed 18 October 2026).

Appendix. The reference library searched

DICTO's reference library (private copies, with public citations in the Kaleidohedra repository's docs/SOURCES.md) holds 254 entries. The full text of the 87 sources below was searched on 19 October 2026 for this paper's subjects: the octet truss and the tetrahedral–octahedral honeycomb; rhombic triacontahedra on the fcc lattice or inside rhombic dodecahedra; interlocking and keyed blocks; connector and glue pieces; tetartoids; cuboctahedra at lattice vertices; golden-section points; and the names Ogdoecontahedron and OG. None contains the kits of this paper. The hits are: the octet truss itself (Fuller 1961) and its part counts (Lalvani 1990); interlocking blocks in the tetroctahedrille (Akpanya et al. 2024); the tetrahedron–octahedron space filler (Lagarias and Zong 2012); Kabai's Golden Octet Truss (a different object, §8) and his triacontahedron–rhombic dodecahedron compound (§8); and the rhombic dodecahedron and triacontahedron as zonohedra (Grünbaum 2010; Senechal 2004; Messer 1995). The other entries (image-only scans and link-only citations of copyrighted books and articles) were not text-searched; no claim is made about them.

Sources checked on the web (18–19 October 2026): searches for rhombic triacontahedra in the octet truss, on the fcc lattice and inscribed in a rhombic dodecahedron; for tetartoids and interlocking glue in the octet honeycomb; Kabai's demonstrations (read in the Wolfram Cloud; descriptions and pictures saved to the library, their code not run); the dimpled-triacontahedron model; Casas (2020), a teaching article whose figures (a triacontahedron assembled from 3D-printed golden rhombohedra; the five golden isozonohedra) were seen but whose text could not be fetched; and, filed in the library on 19 October 2026 as author copies, the woven tiles of Krishnamurthy, Akleman et al. (2020, 2022) and the retessellations of the fcc tiling by Gabbrielli, Jiao and Torquato (2012), none of which places icosahedral solids in the octet.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov	1885	Elements of the Theory of Figures (Nachala ucheniya o figurah)
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
R. Buckminster Fuller	1954	Building construction (geodesic dome)
R. Buckminster Fuller	1961	Synergetic building construction (octet truss)
R. Buckminster Fuller	1962	Tensile-integrity structures
Kenneth D. Snelson	1965	Continuous tension, discontinuous compression structures
Peter J. Pearce	1971	Synthesized natural geometric structures
Stephen C. Baer	1973	Structural system
John P. Hogan	1976	Homohedral construction employing icosahedron
Peter J. Pearce	1976	Minimum inventory maximum diversity building system
Haresh Lalvani	1988	Building structures based on polygonal members and icosahedral symmetry
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
H. Lalvani	1995	Periodic space structures composed of two nodal polyhedra for design applications (US 5,448,868)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
H. Lalvani	1997	Building systems with non-regular polyhedra based on subdivisions of zonohedra (US 5,623,790)
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
S. Kabai	2007	54-Faced Space-Filling Polyhedron (Wolfram Demonstration; description and screenshot)
S. Kabai	2007	Golden Octet Truss (Wolfram Demonstration; description and screenshot)
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2008	A Rhombic Dodecahedron Made of Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
S. Kabai	2008	Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (Wolfram Demonstration; description and screenshot)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra

M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
S. Kabai	2015	Cluster of 14 Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
		Betke Henk 2000 Densest lattice packings of 3 polytopes